Supplementary Material for HCH: History-Calibrated Horizon Repair for Frozen Time-Series Forecasts

This supplement details the data sources, temporal evaluation roles, and frozen base-forecaster configurations, followed by seedwise results, marketwise accuracy, complete ablations, robustness and price-regime diagnostics, and the analytical and implementation details of HCH.

I. Source Resolution, Hourly Products, and Evaluation Origins

All repair experiments use hourly target forecasts even when an upstream price source has a finer sampling interval. Table S1 distinguishes the number of native source rows from the number of records in the selected hourly modeling product and the scored forecast origins. These counts are not interchangeable with the number of statistically independent target windows.

The Shandong 15-minute archive and the selected hourly modeling file cover different date ranges and should not be interpreted as an exact quarter-rate aggregation. The South Australian hourly product contains a final boundary hour; DK1 retains local timestamp and daylight-saving conventions, and the GEFCom source has repeated labels to be resolved by the registered eligible-day reader. The source counts therefore document data extent, not all possible issue-time-legal scored origins. The five Chinese eligible-day train/development/held-out counts are, respectively, Gansu 291/42/83, Shandong 1156/166/330, Shaanxi 284/41/81, Ningxia 100/16/28, and Qinghai 94/14/27. For the external partition, the displayed 5:2:3 is the upper-level host/preparation, post-processing development, and protected-final ratio; further chronological role splits are applied within the first two stages. No protected target is used to compute the additional diagnostic scores reported here.

II. Frozen Base-Forecaster Recipes

The main paper names the four base forecasters without a long hyperparameter catalogue. Table S2 records the frozen recipes used to generate the forecast objects repaired by HCH. These settings are not a shared configuration.

III. Seedwise Variability for the Main Result Tables

The main paper reports ten-seed median MAE/RMSE, sample standard deviations, and rank marks for 44 market–forecaster cells in one unified Table II. Supplementary Tables S3 and S4 present the same frozen values separately for the five development markets and six additional datasets. This separation is for readability and does not define new test cohorts. The cached Base column is deterministic and has no seed deviation. The five displayed methods are Base, δ-Adapter, PIR, UEC-STD, and HCH.

IV. Normalized Accuracy Across Markets and Repairer Size

Table S5 reports a marketwise view of the same 44 market–forecaster MAE cells used in main Table II. Each method's MAE is divided by its matched frozen-Base MAE before the four forecasters are averaged. The values are dimensionless percentages with Base equal to 100, so the table never pools native price-unit errors from different regions. The HCH parameter counts are for the actual market-specific number of admitted forecast-known state channels, not a common 18.7 K configuration.

A. Incremental Network Size and Computation

The ten-seed offline UEC-STD comparison uses its original two-stage `ErrorCorrector` at $T = 24$, input width 3, hidden width 32, and output width 2. It contains $800 + 792 + 128 + 66 = \mathbf{1,786}$ trainable parameters (1.786K). This count covers the trained correction network; seasonal–trend decomposition is a nonparametric preprocessing operation whose CPU cost is not represented by the weight count. HCH's market-specific direction network contains 13.336–20.248K parameters, while the profiled PIR and δ-Adapter settings contain 190.5K and 289.8K. Thus HCH is smaller than the profiled PIR/δ configurations but not smaller than UEC-STD. The normalized accuracy comparison and the learned-parameter comparison answer distinct questions.

V. Full Component Ablation and the Direction–Calibration Interaction

Main Table III reports the frozen Base, full HCH, and four single-intervention variants on the same 24 configurations, with five-seed means first normalized to the corresponding Base and then averaged equally across cells. Supplementary Table S6 retains the disaggregated 24-cell comparison, expressed instead as the percentage reduction in MAE achieved by full HCH relative to each control. These two presentations have different denominators and should not be read as numerically identical columns. The four registered interventions change direction estimation, distance calibration, historical

TABLE S1
SOURCE VERSUS MODELING RECORD COUNTS AND EVALUATION-ORIGIN COUNTS FOR THE FROZEN HCH DATA INVENTORY.

Dataset	Native cadence	Native rows	Hourly product rows	Exog.	Scored origins	Top-level role
Gansu	1 h	14,112	14,112	7	42	Domestic development
Shandong	15 min	160,800	39,816	9	166	Domestic development
Shaanxi	1 h	10,752	10,752	8	41	Domestic development
Ningxia	1 h	4,224	4,224	7	16	Domestic development
Qinghai	1 h	4,440	4,440	4	14	Domestic development
Germany	1 h	52,416	52,416	2	109	External development
PJM	1 h	52,416	52,416	1	109	External development
GEFCom2014	1 h	25,968	25,968	2	53	External development
Nord Pool NP	1 h	52,416	52,416	2	108	External development
Nord Pool DK1	1 h	24,189	24,189	0	289	Frozen-HCH confirmation
NEM-SA1	5 min	105,408	8,785	0	108	Frozen-HCH confirmation

TABLE S2
FROZEN BASE-FORECASTER RECIPES USED BY THE HCH BENCHMARK.

Base forecaster	Epochs	Batch	Learning rate	Scheduler	Architecture settings
PatchTST	100	128	10^{-4}	OneCycleLR	$d_{\text{model}} = 512$, 8 heads, 2 encoder layers, FFN 2048, dropout 0.05, patch length 16, stride 8.
TimeMixer	10	128	10^{-2}	OneCycleLR	$d_{\text{model}} = 16$, $d_{\text{ff}} = 32$, 2 layers, dropout 0.1, 3 down-sampling layers, window 2, average down-sampling, moving average 25.
iTransformer	10	32	10^{-4}	none	$d_{\text{model}} = 256$, $d_{\text{ff}} = 256$, 8 heads, 2 encoder layers, dropout 0.1, GELU; frozen recipe seed 2023.
LSTM	100	128	10^{-3}	none	hidden width 64, 2 recurrent layers, dropout 0.1; frozen recipe seed 7.

All four use a 168-step input window, 24-step forecast horizon, one target channel, Adam, and zero weight decay. PatchTST and TimeMixer use OneCycleLR with $\texttt{pct_start} = 0.3$; iTransformer and LSTM use no scheduler.

TABLE S3
FIVE CHINESE MARKETS, FOUR FROZEN BASE FORECASTERS, TEN SEEDS: MEDIAN MAE / RMSE; BOLD BEST, UNDERLINE SECOND BEST.

Backbone	Base	δ-Adapter	PIR	UEC-STD	HCH
Gansu LSTM	80.661 / 99.715	80.588 ± 0.62 / 99.743 ± 0.49	77.642 ± 0.41 / 99.367 ± 1.17	80.708 ± 0.44 / 99.740 ± 0.39	**69.799 ± 2.54 / 89.243 ± 2.98**
PatchTST	77.104 / 97.760	77.657 ± 0.37 / 98.095 ± 0.32	77.337 ± 0.99 / 98.651 ± 2.20	77.060 ± 0.21 / 97.688 ± 0.23	**69.889 ± 0.92 / 90.095 ± 1.11**
TimeMixer	76.607 / 97.888	76.758 ± 0.28 / 97.977 ± 0.21	87.885 ± 3.37 / 112.380 ± 4.45	76.652 ± 0.29 / 97.943 ± 0.25	**71.992 ± 1.26 / 93.519 ± 1.56**
iTransformer	79.174 / 101.339	79.259 ± 0.13 / 101.380 ± 0.23	75.959 ± 0.98 / 98.401 ± 1.32	79.156 ± 0.22 / 101.291 ± 0.23	**71.771 ± 0.97 / 94.353 ± 1.42**
Shandong LSTM	84.711 / 114.386	84.967 ± 0.35 / 114.618 ± 0.36	85.539 ± 1.00 / 115.261 ± 0.90	84.649 ± 0.73 / 114.440 ± 0.88	**74.188 ± 1.61 / 102.548 ± 1.99**
PatchTST	76.867 / 105.616	78.042 ± 0.21 / 106.210 ± 0.21	83.717 ± 1.15 / 110.991 ± 1.09	76.773 ± 0.29 / 105.593 ± 0.22	**74.719 ± 0.97 / 101.536 ± 1.33**
TimeMixer	78.992 / 108.428	78.837 ± 0.37 / 108.200 ± 0.31	87.794 ± 1.24 / 115.596 ± 1.46	78.983 ± 0.51 / 108.413 ± 0.40	**74.053 ± 1.04 / 102.093 ± 1.18**
iTransformer	75.070 / 105.517	75.663 ± 0.17 / 105.448 ± 0.13	81.604 ± 2.30 / 109.415 ± 2.50	75.159 ± 0.14 / 105.577 ± 0.15	**72.365 ± 1.50 / 99.937 ± 1.91**
Shaanxi LSTM	109.456 / 157.116	109.578 ± 0.19 / 157.248 ± 0.23	114.267 ± 7.24 / 166.270 ± 15.53	109.567 ± 0.57 / 157.169 ± 0.69	**98.110 ± 2.57 / 144.113 ± 2.55**
PatchTST	111.965 / 159.547	111.987 ± 0.19 / 159.445 ± 0.19	123.212 ± 5.71 / 173.542 ± 6.37	112.012 ± 0.15 / 159.615 ± 0.28	**108.869 ± 1.74 / 153.978 ± 2.01**
TimeMixer	103.592 / 150.420	103.719 ± 0.21 / 150.309 ± 0.25	118.735 ± 5.98 / 167.622 ± 6.89	103.638 ± 0.13 / 150.512 ± 0.21	**97.524 ± 0.56 / 143.401 ± 1.30**
iTransformer	112.507 / 160.605	112.497 ± 0.14 / 160.650 ± 0.21	122.647 ± 7.64 / 171.124 ± 8.35	112.508 ± 0.10 / 160.675 ± 0.23	**106.488 ± 0.92 / 153.674 ± 1.25**
Ningxia LSTM	70.222 / 93.089	70.210 ± 0.23 / 93.144 ± 0.23	78.460 ± 5.78 / 106.724 ± 5.64	70.175 ± 0.33 / 93.085 ± 0.50	**61.761 ± 5.12 / 78.740 ± 5.35**
PatchTST	65.074 / 82.300	64.958 ± 0.19 / 82.229 ± 0.09	77.262 ± 4.21 / 103.274 ± 6.62	65.002 ± 0.20 / 82.229 ± 0.14	**55.978 ± 1.85 / 73.074 ± 2.50**
TimeMixer	78.347 / 96.517	78.219 ± 0.47 / 96.453 ± 0.42	78.904 ± 3.75 / 102.871 ± 5.68	78.289 ± 0.39 / 96.486 ± 0.20	**56.925 ± 3.71 / 71.938 ± 5.58**
iTransformer	71.314 / 89.881	70.548 ± 0.59 / 89.948 ± 0.37	76.616 ± 4.58 / 100.245 ± 5.50	71.315 ± 0.30 / 89.893 ± 0.26	**57.868 ± 3.31 / 73.718 ± 4.04**
Qinghai LSTM	171.295 / 202.040	170.814 ± 0.78 / 201.616 ± 0.68	**128.117 ± 7.30 / 171.378 ± 5.79**	171.425 ± 0.80 / 202.401 ± 0.92	150.494 ± 6.11 / 195.619 ± 8.46
PatchTST	143.256 / 185.193	143.188 ± 0.63 / 185.896 ± 0.67	154.334 ± 3.83 / 210.891 ± 7.04	143.346 ± 0.39 / 185.320 ± 0.45	**141.329 ± 0.93 / 182.244 ± 1.23**
TimeMixer	157.571 / 206.163	157.785 ± 0.57 / 206.296 ± 1.04	**150.371 ± 3.74** / 213.120 ± 6.75	157.541 ± 0.85 / 206.091 ± 0.85	153.274 ± 2.06 / **200.807 ± 2.88**
iTransformer	161.447 / 205.684	160.784 ± 0.56 / 205.516 ± 0.61	**143.227 ± 1.86 / 196.866 ± 3.67**	161.498 ± 0.49 / 205.788 ± 0.67	161.061 ± 0.32 / 205.418 ± 0.34

scale, and the chronological fitting procedure, respectively. The paired two-direction calibration analysis is reported only in this Supplement. Main Table IV instead compares normalized accuracy and incremental repairer size.

A. One Combined Intervention: Direction Replacement and Fixed Distance

To measure whether calibration has a direction-estimator-specific effect, we complete one two-factor experiment on the already registered development data. The direction factor uses either the full history-conditioned GRU estimator or the registered current-context direction control. The distance factor uses either each direction's chronological OOF-calibrated coefficient or the same fixed coefficient $\alpha = 1$. No new model was fitted: the previously unreported fourth cell was calculated from cached signed directions, the matched historical scale, and the same Base forecast and targets.

The relevant quantities are $\mathrm{MAE}(H, \hat{\alpha}_H)$, $\mathrm{MAE}(H, 1)$, $\mathrm{MAE}(C, \hat{\alpha}_C)$ and $\mathrm{MAE}(C, 1)$. We measure the calibration gain for each direction, normalized to the same Base MAE, and then compare those gains; positive values mean the calibrated coefficient reduces error relative to the fixed distance.

These matched results show that chronological scalar distance calibration can improve both directional implementations. They do not require the directional and distance estimators to be mathematically inseparable, and they do not

TABLE S4
EXTERNAL TRANSFER, SIX MARKETS, TEN SEEDS: MEDIAN MAE / RMSE; BOLD BEST, UNDERLINE SECOND BEST.

Backbone	Base	δ-Adapter	PIR	UEC-STD	HCH
A. Untouched external confirmation					
Nord Pool DK1 PatchTST	21.128 / 28.620	20.681 ± 0.09 / 27.995 ± 0.10	20.907 ± 0.46 / 28.222 ± 0.50	20.566 ± 0.26 / 27.909 ± 0.30	**20.482 ± 0.13 / 27.825 ± 0.16**
TimeMixer	20.087 / 27.046	20.102 ± 0.04 / 27.042 ± 0.04	21.066 ± 0.57 / 28.206 ± 0.66	19.716 ± 0.12 / 26.597 ± 0.14	**19.558 ± 0.11 / 26.364 ± 0.11**
iTransformer	20.782 / 27.931	20.529 ± 0.04 / 27.587 ± 0.05	21.734 ± 0.26 / 29.061 ± 0.38	20.473 ± 0.13 / 27.563 ± 0.16	**20.412 ± 0.04 / 27.435 ± 0.06**
LSTM	30.850 / 40.026	30.868 ± 0.18 / 40.033 ± 0.17	**21.555 ± 0.23 / 28.853 ± 0.29**	25.660 ± 0.82 / 33.369 ± 0.76	24.094 ± 1.01 / 31.919 ± 1.21
Australian NEM South Australia PatchTST	130.488 / 430.218	130.602 ± 0.31 / 430.000 ± 0.29	133.675 ± 4.90 / 444.227 ± 6.20	**126.411 ± 3.12 / 420.899 ± 3.65**	128.541 ± 0.51 / 427.018 ± 0.86
TimeMixer	92.558 / 390.038	93.574 ± 0.75 / 390.216 ± 0.32	122.477 ± 5.45 / 437.496 ± 8.23	91.529 ± 0.71 / 389.382 ± 0.53	**90.379 ± 0.32 / 388.037 ± 0.16**
iTransformer	120.342 / 416.345	120.620 ± 0.43 / 416.302 ± 0.16	157.397 ± 4.45 / 494.530 ± 11.97	**118.422 ± 1.22 / 412.926 ± 1.66**	119.033 ± 0.46 / 413.998 ± 0.70
LSTM	78.529 / 378.012	79.152 ± 0.81 / 378.069 ± 0.21	119.033 ± 2.30 / 438.084 ± 5.54	**77.464 ± 2.11** / 378.017 ± 0.26	77.485 ± 0.35 / **377.787 ± 0.12**
B. External development breadth					
Germany PatchTST	4.757 / 6.397	4.743 ± 0.01 / 6.398 ± 0.02	5.073 ± 0.11 / 6.813 ± 0.13	4.738 ± 0.02 / 6.412 ± 0.02	**3.926 ± 0.03 / 5.490 ± 0.05**
TimeMixer	4.707 / 6.385	4.716 ± 0.01 / 6.394 ± 0.01	5.121 ± 0.12 / 6.751 ± 0.14	4.692 ± 0.01 / 6.384 ± 0.01	**3.952 ± 0.04 / 5.471 ± 0.06**
iTransformer	4.734 / 6.486	4.719 ± 0.01 / 6.475 ± 0.02	5.056 ± 0.14 / 6.676 ± 0.19	4.694 ± 0.01 / 6.467 ± 0.02	**3.988 ± 0.04 / 5.607 ± 0.04**
LSTM	6.917 / 8.720	6.851 ± 0.05 / 8.659 ± 0.05	5.376 ± 0.05 / 7.184 ± 0.05	6.560 ± 0.08 / 8.374 ± 0.07	**4.423 ± 0.05 / 5.825 ± 0.06**
PJM PatchTST	3.020 / 4.359	2.974 ± 0.02 / 4.316 ± 0.03	3.273 ± 0.04 / 4.607 ± 0.05	2.973 ± 0.02 / 4.307 ± 0.02	**2.922 ± 0.03 / 4.177 ± 0.04**
TimeMixer	3.221 / 4.579	3.237 ± 0.02 / 4.588 ± 0.02	3.154 ± 0.07 / 4.515 ± 0.08	3.136 ± 0.02 / 4.432 ± 0.04	**2.827 ± 0.03 / 3.974 ± 0.04**
iTransformer	3.303 / 4.625	3.195 ± 0.01 / 4.528 ± 0.01	**2.925 ± 0.04 / 4.213 ± 0.06**	3.298 ± 0.02 / 4.601 ± 0.01	2.976 ± 0.04 / 4.224 ± 0.06
LSTM	4.283 / 5.810	4.203 ± 0.03 / 5.702 ± 0.04	**3.323 ± 0.09 / 4.665 ± 0.10**	3.921 ± 0.14 / 5.382 ± 0.16	3.420 ± 0.10 / 4.928 ± 0.10
GEFCom2014 PatchTST	**7.405 / 13.171**	7.441 ± 0.03 / 13.344 ± 0.07	9.996 ± 0.20 / 16.515 ± 0.25	7.536 ± 0.08 / 13.735 ± 0.15	7.626 ± 0.08 / 13.621 ± 0.09
TimeMixer	8.863 / 14.029	8.852 ± 0.04 / 14.028 ± 0.02	10.404 ± 0.24 / 17.231 ± 0.34	**8.654 ± 0.07 / 13.969 ± 0.08**	8.924 ± 0.13 / 14.273 ± 0.17
iTransformer	**7.578** / 13.496	7.590 ± 0.00 / 13.518 ± 0.02	9.501 ± 0.22 / 14.996 ± 0.29	7.746 ± 0.07 / **13.370 ± 0.15**	7.854 ± 0.09 / 13.634 ± 0.07
LSTM	**8.298 / 13.272**	8.301 ± 0.01 / 13.279 ± 0.02	10.536 ± 0.21 / 16.794 ± 0.21	8.299 ± 0.05 / 13.275 ± 0.20	8.462 ± 0.24 / 13.387 ± 0.28
Nord Pool (benchmark) PatchTST	1.979 / 3.474	1.985 ± 0.01 / 3.470 ± 0.01	2.109 ± 0.06 / 3.396 ± 0.06	2.003 ± 0.01 / 3.475 ± 0.01	**1.907 ± 0.03 / 3.378 ± 0.03**
TimeMixer	2.002 / 3.471	2.010 ± 0.01 / 3.479 ± 0.01	2.331 ± 0.04 / 3.607 ± 0.03	2.013 ± 0.02 / 3.463 ± 0.02	**1.967 ± 0.03 / 3.404 ± 0.03**
iTransformer	2.120 / 3.478	2.132 ± 0.02 / 3.486 ± 0.01	2.507 ± 0.06 / 3.989 ± 0.10	2.095 ± 0.02 / 3.460 ± 0.02	**1.983 ± 0.01 / 3.359 ± 0.02**
LSTM	1.990 / 3.363	2.060 ± 0.04 / 3.386 ± 0.02	2.204 ± 0.04 / 3.625 ± 0.02	1.998 ± 0.01 / 3.363 ± 0.00	**1.769 ± 0.02 / 3.123 ± 0.03**

TABLE S5
MARKETWISE MEAN NORMALIZED MAE OVER FOUR FROZEN FORECASTERS AND INCREMENTAL HCH REPAIR-NETWORK SIZE. LOWER RELATIVE MAE IS BETTER; ALL ROWS HAVE BASE= 100.

Market	Role	δ-Adapter	PIR	UEC-STD	HCH	HCH params (K)
Gansu	Domestic development	100.23	101.81	100.01	**90.45**	18.712
Shandong	Domestic development	100.61	107.43	99.98	**93.73**	20.248
Shaanxi	Domestic development	100.06	109.52	100.05	**93.92**	19.480
Ningxia	Domestic development	99.64	109.65	99.94	**81.94**	18.712
Qinghai	Domestic development	99.85	**91.67**	100.04	95.89	16.408
Germany	External development	99.66	99.99	98.32	**78.67**	14.872
PJM	External development	98.46	93.11	96.80	**88.62**	14.104
GEFCom2014	External development	**100.14**	126.18	100.41	102.32	14.872
Nord Pool NP	External development	101.20	113.00	100.25	**94.26**	14.872
Nord Pool DK1	Frozen-HCH confirmation	99.20	94.57	94.30	**92.66**	13.336
NEM South Australia	Frozen-HCH confirmation	100.55	129.28	**98.20**	98.43	13.336
Equal-weight mean (44 cells)	—	99.96	106.93	98.94	**91.90**	13.336–20.248

Scores are computed from the displayed ten-seed summary MAEs in main Table II. The calculation is 100 $\mathrm{mean}_h[\mathrm{MAE}_{m,h}/\mathrm{MAE}_{\mathrm{Base},h}]$ within each market, then averaged over the 44 cells. Ranking marks apply only to the displayed marketwise normalized MAE and do not assert statistical significance.

isolate GRU architecture alone: the current-context control also omits the seven historical patches. The calculation replays 120 development forecast caches (24 cells × 5 seeds), with exact timestamp, frozen-Base and outcome identity across the control and corresponding full-model files. The original control predictions reconstruct to a maximum absolute discrepancy of 3.72×10^{-5}. The registered main predictions and the original four single-intervention figures are unchanged.

VI. TEMPORAL AND MISSING-HISTORY ROBUSTNESS

A. Chronology Robustness

Whole-window averages can hide sustained local failures. Figure S3 therefore reports the registered sliding 14-day chronology diagnostic at a readable 2×2 scale. The marked interval uses the frozen event-box semantics: it is the longest run for which at least one base forecaster has all of its evaluated HCH seeds below zero. Overlapping windows describe persistence rather than independent failure events.

Table S8 completes the markets omitted from the four-panel chronology figure using the same statistic and source rows. Among these five markets, only Gansu contains negative median windows; the worst window remains positive in Shaanxi, Ningxia, Qinghai, and PJM.

B. Missing Historical Residuals

Figure S4 shows the registered history-residual masking intervention. Extra masking of 10%, 30%, or 50% is applied to the revealed residual stream at issuance while the corrector remains frozen; 0% extra masking is the paired reference. The five Chinese development markets and four frozen forecasters contribute 20 cells, with three registered seeds per cell. At 50% masking, 19/20 cells still improve on Base and the remaining cell is level with Base, but 12/20 cells lose ground relative to their own unmasked HCH outcomes. Let $g_i(p)$ denote the Base-relative percentage-point gain of cell i at mask level p. The plotted aggregate change is

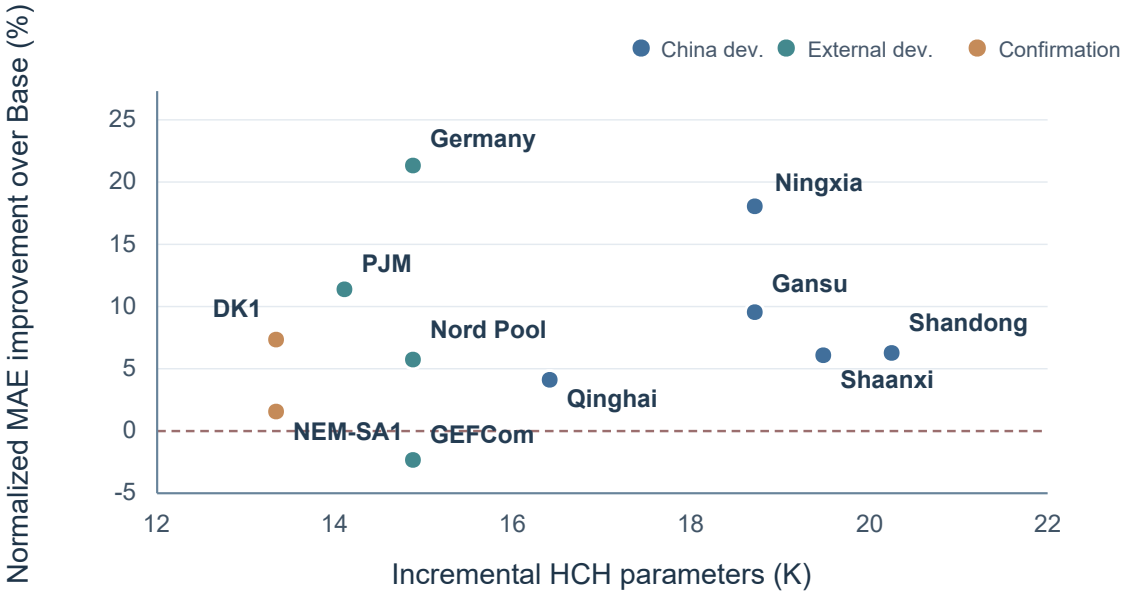


Fig. S1. Marketwise accuracy–size distribution for the reference HCH plug-in. Each point averages the four matched Base-normalized MAE changes within the indicated market, using ten-seed summary results from main Table II. Horizontal coordinates are the exact incremental direction-network parameter counts for the market-specific forecast-known covariate inventory. Colors distinguish the original development and confirmation roles; the scatter is descriptive and does not attribute changes in forecasting accuracy to parameter count.

TABLE S6

MATCHED FIVE-SEED ABLATION RESULTS ON 24 MARKET–FORECASTER CELLS. ENTRIES GIVE THE PERCENTAGE MAE IMPROVEMENT OF FULL HCH OVER EACH REGISTERED CONTROL (POSITIVE FAVORS HCH). EACH COLUMN MODIFIES ONE REGISTERED PROCEDURE; COMBINED REMOVALS ARE NOT INCLUDED.

Market	Frozen Base	Direction ctrl.	$\alpha = 1$	Fixed scale	No CACC
Gansu	PatchTST	-0.22	+3.43	+0.74	+6.57
	TimeMixer	-0.29	+1.63	+1.02	+4.53
	iTransformer	+1.09	+3.35	+1.15	+7.46
	LSTM	+5.49	+6.78	-0.26	+9.86
Shandong	PatchTST	+0.12	-1.26	+1.62	-0.05
	TimeMixer	+0.45	+0.58	+1.94	+0.57
	iTransformer	+0.39	-1.33	+3.16	-3.69
	LSTM	+1.09	+4.59	+0.38	+1.57
Shaanxi	PatchTST	-1.32	-0.13	+0.32	+1.28
	TimeMixer	-0.64	+1.10	+1.53	+3.51
	iTransformer	-0.38	+0.56	+1.92	+3.81
	LSTM	+0.27	+3.52	+2.63	+5.63
Ningxia	PatchTST	+1.69	+5.20	+0.22	+11.83
	TimeMixer	+7.33	+17.79	+6.87	+18.29
	iTransformer	-1.61	+8.31	-0.67	+9.06
	LSTM	-1.85	+3.02	-0.27	+9.70
Qinghai	PatchTST	+0.34	-0.63	+0.84	+0.46
	TimeMixer	+2.15	-0.26	+0.89	+0.25
	iTransformer	+0.19	+0.23	+0.09	+0.06
	LSTM	+12.15	+7.50	-0.17	+2.85
Germany	PatchTST	-2.29	+9.18	+0.30	+8.03
	TimeMixer	-2.29	+8.12	+0.15	+7.50
PJM	PatchTST	+1.07	+0.18	-0.27	+3.95
	TimeMixer	-1.77	+6.15	-1.45	+8.24

$\text{median}_i\, g_i(50) - \text{median}_i\, g_i(0) \simeq +0.480$, while the paired statistic is $\text{median}_i\{g_i(50) - g_i(0)\} \simeq -0.276$. Both use the same cellwise gain definition; the order of subtraction and median aggregation differs, so the signs need not agree. These results do not show that additional masking universally improves forecasting accuracy.

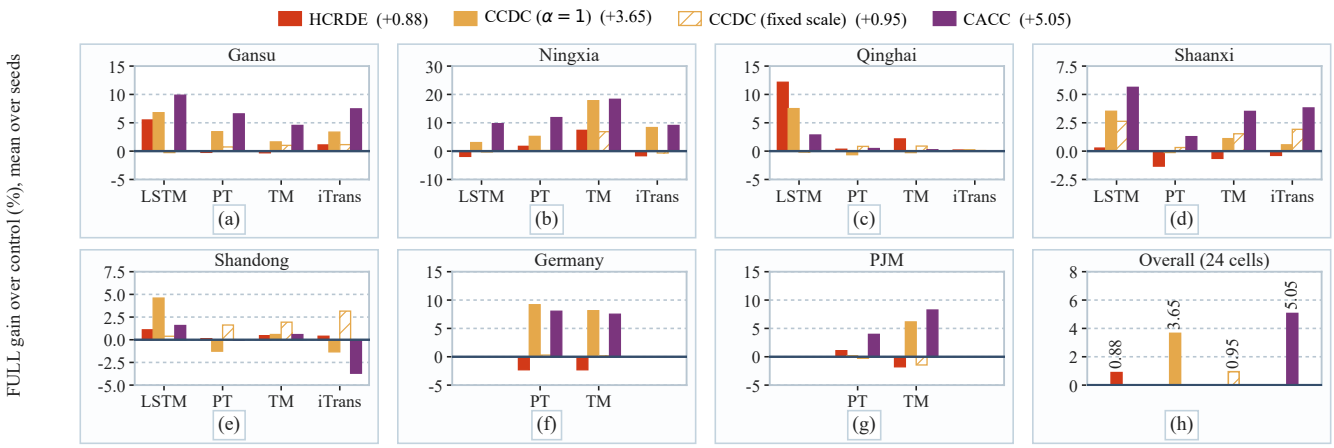


Fig. S2. Four registered single-intervention HCH ablations. This figure visualizes the full-cell results of Table S6, and is kept in the Supplement to avoid repeating the same comparisons in the main paper.

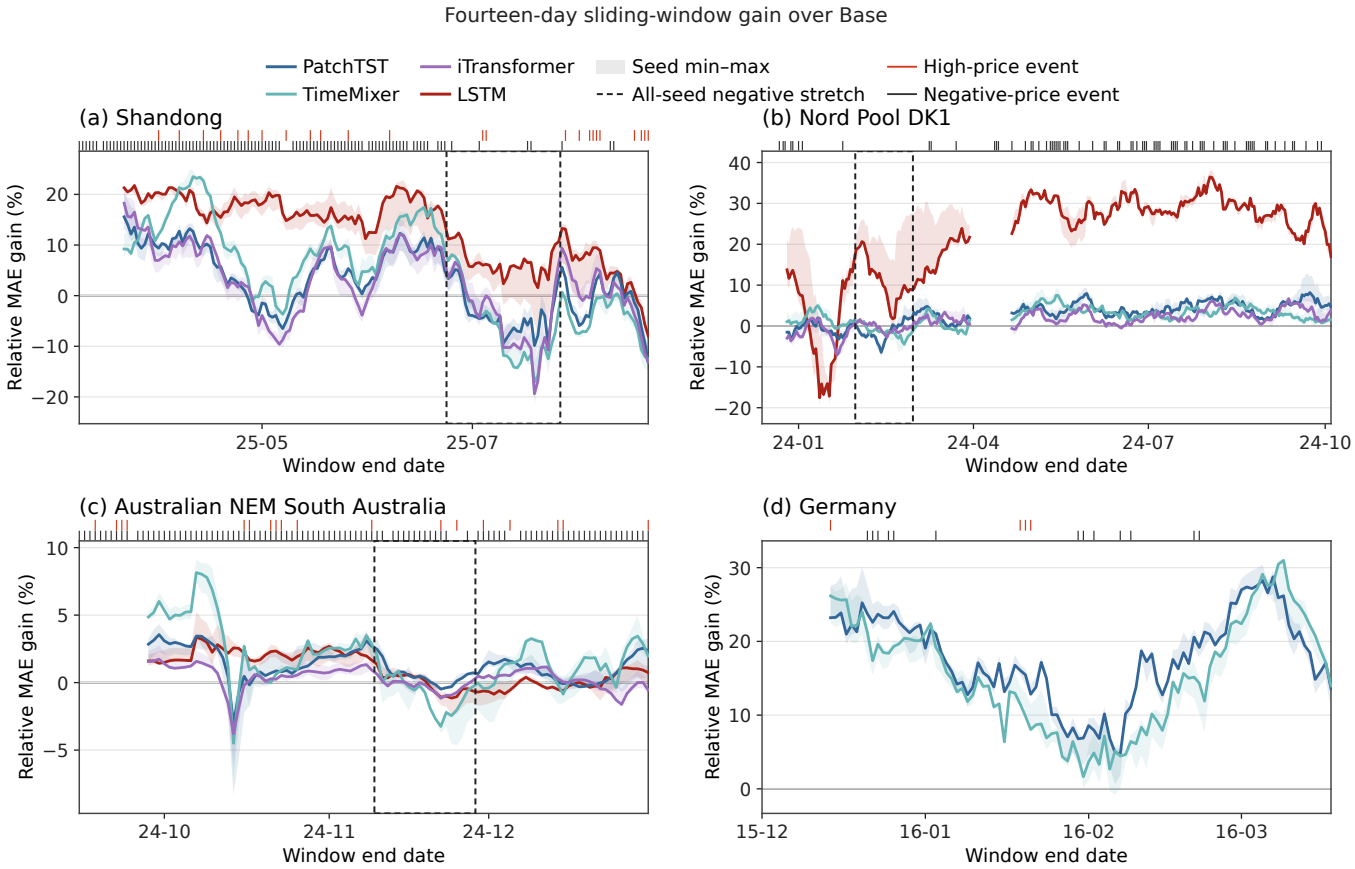


Fig. S3. Chronology robustness under the registered sliding 14-day statistic. Local negative periods can coexist with positive whole-window performance; the highlighted runs follow the frozen event-box definition above.

VII. Regime-Specific Prediction and Repair Diagnostics

Figure S5 provides the native-unit, matched five-method development evidence underlying the conditional comparisons. Each bar uses the same issued Base, target time and truth within a market, and averages over four forecasters and ten registered seeds. The six panels retain the original market-level absolute MAE values, rather than combining currencies. The main paper's high-price panel averages two within-

TABLE S7

COMBINED DIRECTION–DISTANCE CONTROL. EACH ROW SUMMARIZES FIVE REGISTERED DEVELOPMENT SEEDS PER MARKET–FORECASTER CELL; G_H AND G_C ARE NORMALIZED CALIBRATION GAINS IN BASE-MAE PERCENTAGE POINTS, NOT POOLED NATIVE ERRORS.

Market	G_H	G_C	$G_H - G_C$
Gansu	3.548	2.613	+0.935
Germany	7.912	9.195	-1.283
PJM	2.943	3.494	-0.551
Ningxia	7.668	6.933	+0.735
Qinghai	1.601	-0.974	+2.575
Shaanxi	1.179	1.337	-0.158
Shandong	0.575	0.345	+0.230
24-cell mean	3.333	2.766	+0.566

Calibration improves 19/24 cells for either direction estimator. In 12/24 cells the calibrated HCRDE benefits more; the 24-cell median interaction is +0.035 Base-MAE percentage points.

market relative MAE ratios, preventing different native price units from being pooled. Its large zero-crossing panel is intentionally narrower than the broad cross-zero panel shown here: it additionally requires adjacent absolute change above Shandong's TRAIN-only 95th percentile (119 rather than 225 eligible target hours).

Figure S6 decomposes the Shandong negative-hour corrections within the same issue-time cohort. The improvement indicator requires the signed adjustment to reduce absolute error, not merely to point toward the truth. HCH has the largest beneficial-adjustment fraction (70.8%) but a nonminimal overshoot fraction (15.4%); its net MAE advantage is not an overshoot-protection guarantee.

Negative-price cross-market evidence. The main paper reports the *same-cohort* five-method Shandong negative-hour comparison (four forecasters, ten seeds). Earlier three-seed HCH/Base negative-hour summaries for DK1 and South Australia are retained as historical evidence only: they use different windows and confirmation roles, and do not support matched five-method ranking on their negative-price hours. We do not combine those archives with the Chinese ten-seed development estimates. All conditional figures are outcome-defined retrospective diagnostics, not independent newly held-out experiments.

TABLE S8

THE FIVE MARKETS NOT DRAWN IN THE CHRONOLOGY FIGURE, AT THE IDENTICAL STATISTIC AND FROM THE SAME PLOT-READY ROWS; WITH THE FOUR DRAWN PANELS THEY COMPLETE THE REGISTERED POPULATION. ALL ARE VALIDATION-SPLIT EXCEPT PJM, WHICH IS DEVELOPMENT-SPLIT; WORST AND BEST ARE A MARKET'S EXTREME PER-WINDOW REPAIR GAINS.

Market	Base fcst.	Win.	Cov/gap (days)	Worst (%)	Best (%)	Neg. med.	No seed > 0
Gansu	4	28	41 / 0	−0.25	+19.67	2	0
Shaanxi	4	15	49 / 21	+0.59	+16.72	0	0
Ningxia	4	3	16 / 0	+2.30	+33.17	0	0
Qinghai	4	1	14 / 0	+0.10	+13.71	0	0
PJM	2	96	109 / 0	+0.94	+28.17	0	0

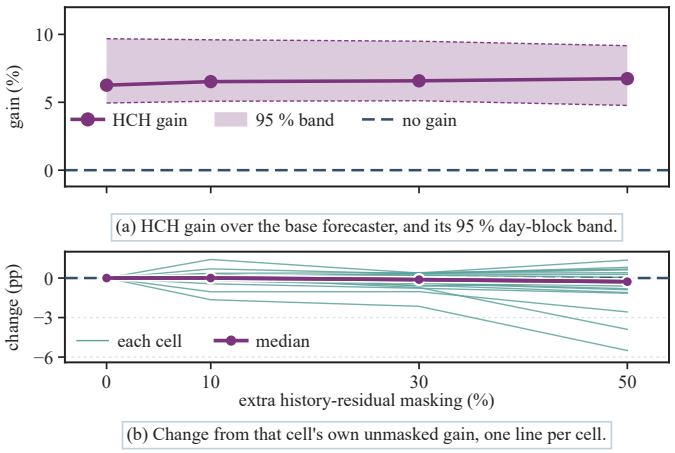


Fig. S4. History-residual masking at issue time. (a) Median Base-relative MAE gain over 20 cells, with day-block resampling bands conditional on the registered markets and their distinct evaluation windows; (b) paired change relative to unmasked HCH within each cell. Three seeds per cell are summarized before cross-cell aggregation. The two statistics need not agree.

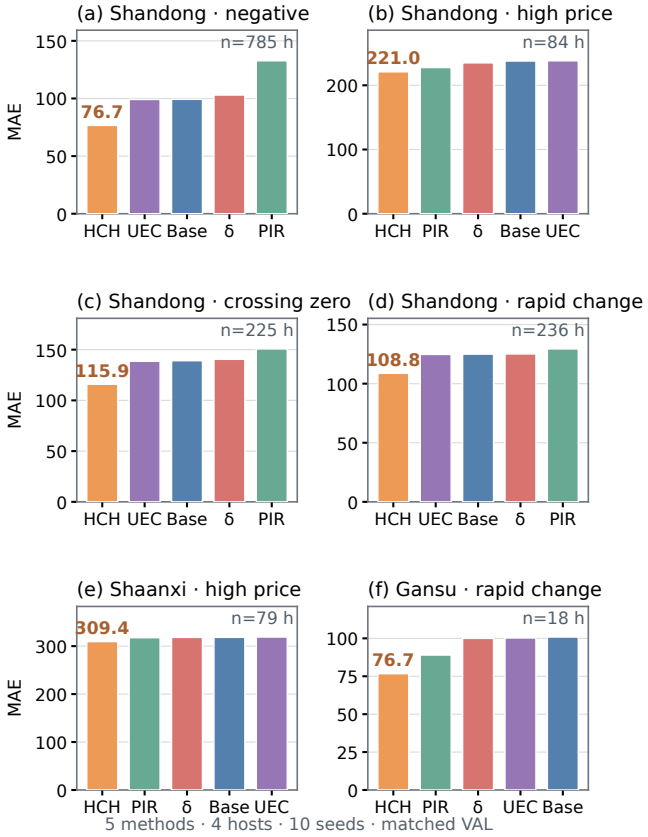


Fig. S5. Five-method native-MAE comparisons on source-matched ten-seed development records: Shandong negative-price, high-price, cross-zero and rapid-change hours, Shaanxi high-price hours, and Gansu rapid-change hours. The Gansu jump subset has only 18 scored hours and is exploratory. Regimes are selected after outcomes are revealed and cannot be supplied at issuance.

A. *Within-Day Extremum Timing on Shandong*

Beyond aggregate MAE, the historical-repair record contains a source-registered diagnostic of phase error for the Shandong develop-

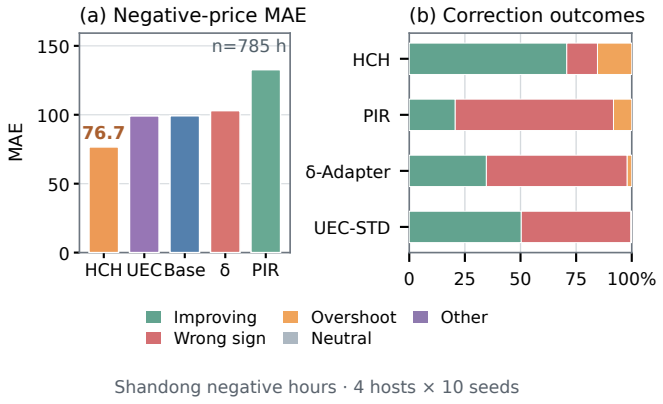


Fig. S6. Shandong negative-price development diagnostics on matched four-forecaster ten-seed outputs: (a) native MAE of Base and four post-processors; (b) correcting, wrong-sign, overshooting, neutral, and other adjustment shares. The Base has no correction categories.

ment market. Fig. S7 compares the frozen Base and HCH on the same 166 validation origins for each of four frozen forecasters. The normal-day statistic averages circular peak and trough timing distances over 24 hours; the extreme-event statistic gives equal weight to the high-price peak and negative-price trough subsets. The high-price threshold (608.57127 in native price units) was estimated on the training period. The subsets comprise 24 high-price days, 106 negative-price days, 11 overlapping days, and 47 normal days. Each bar is the median phase-delay improvement over seeds 7, 17, and 37; whiskers show seed minima and maxima, not confidence intervals.

The diagnostic complements the population-level profile geometry analysis: it directly measures timing displacement rather than correlation or dispersion. Phase improvement varies across forecasters and regimes. In particular, the TimeMixer extreme-event median is negative, whereas the PatchTST and LSTM medians are positive in both regimes. The panel is a descriptive, private-market development analysis and is not treated as an additional public-market confirmation.

VIII. Analytical Notes on Direction and Distance

This section is analysis only; it does not add an operation to HCH. It retains only dimensionally consistent interpretations of the frozen repair chain.

A. One Repair Mapping and Two Estimation Stages

An issued forecast is modified by one vector-valued repair rule,

$$\widehat{\mathbf{y}}_t^{\,r} = \widehat{\mathbf{y}}_t^{\,b} + f_\phi\left(\widehat{\mathbf{y}}_t^{\,b}, \mathcal{E}_t\right), \qquad f_\phi = s_t \widehat{\alpha}\, \mathbf{u}_{\theta^*,t}. \tag{S1}$$

The directional estimator and the scalar distance estimator are two historical-fitting stages of this single method, not a claim that the same correction vector cannot be output by a monolithic network. The separation specifies how each estimate is obtained from available historical evidence: a normalized horizon profile is learned by the compact reference encoder, and a shared amplitude is calibrated from its chronological out-of-fit proposals. This is an estimation-and-execution choice, rather than a mathematical impossibility result about other architectures. At issuance, its deployment artifact consists of the fitted direction weights, one nonnegative calibration coefficient and the deterministic history-based scale.

B. Weighted-Median Calibration in Normalized Absolute Loss

For an eligible chronological out-of-fold coordinate (i,h), let $e_{i,h}$ denote the normalized residual target used by CCDC and let $u_{i,h}$ denote the corresponding out-of-fold direction coordinate. The implementation keeps finite coordinates with $|u_{i,h}| > 10^{-12}$. Define

$$z_{i,h} = \frac{e_{i,h}}{u_{i,h}}, \qquad w_{i,h} = |u_{i,h}|. \tag{S2}$$

For any nonnegative scalar α,

$$|e_{i,h} - \alpha u_{i,h}| = w_{i,h}|z_{i,h} - \alpha|. \tag{S3}$$

Hence the pooled normalized absolute-error objective is

$$Q(\alpha) = \sum_{(i,h)\in\mathcal{E}} w_{i,h}|z_{i,h} - \alpha|, \qquad \alpha \geq 0, \tag{S4}$$

where $\mathcal{E}$ is the finite eligible OOF coordinate set. The unconstrained minimizers are weighted medians of $z_{i,h}$; imposing the implemented nonnegative constraint gives

$$\widehat{\alpha} = \max\left\{0, \operatorname{wmed}_{(i,h)\in\mathcal{E}}\left(z_{i,h}; w_{i,h}\right)\right\}. \tag{S5}$$

Negative ratios remain admissible; only the final scalar is projected onto the nonnegative half-line. This statement is dimensionless throughout and matches the frozen per-market–base-forecaster CCDC operation.

C. A Dimensionally Consistent Direction–Distance Decomposition

Let $r \in \mathbb{R}^H$ be a realized raw-price residual and $c \in \mathbb{R}^H$ a raw-price correction for the same horizon. For nonzero vectors, define

$$A = \|r\|_2, \qquad B = \|c\|_2, \qquad p = \frac{r}{A}, \qquad q = \frac{c}{B}. \tag{S6}$$

Then $\|p\|_2 = \|q\|_2 = 1$ and

$$\|r - c\|_2^2 = (A - B)^2 + AB\,\|p - q\|_2^2. \tag{S7}$$

The first term measures magnitude mismatch and the second directional disagreement, scaled by both magnitudes. The two effects are coupled rather than assigned exclusively to separate HCH components. If $A = 0$ or $B = 0$, the expression extends continuously because the product term vanishes.

Writing

$$D = (A - B)^2 + AB\,\|p - q\|_2^2, \tag{S8}$$

standard norm equivalence gives

$$\frac{1}{H}\sqrt{D} \;\leq\; \frac{1}{H}\|r - c\|_1 \;\leq\; \frac{1}{\sqrt{H}}\sqrt{D}. \tag{S9}$$

All terms are in raw-price units. These equations are analytical interpretations only; HCH does not optimize, calibrate, route, or deploy through them.

IX. Chronology-Aware Capacity Selection Procedure

The main paper gives the compact mathematical definition of CACC. The complete frozen procedure is:

1) Start from a legal chronological fitting prefix $\mathcal{P}$. Set
$$n_{\text{val}} = \max(5, \lceil 0.20|\mathcal{P}|\rceil). \tag{S10}$$
The latest n_{val} days form $\mathcal{P}_{\text{val}}$ and the preceding days form $\mathcal{P}_{\text{tr}}$; at least 15 inner-training days are required.
2) From one fixed initialization θ_0, evaluate direction loss at step 0 and after every 25 full-batch updates through step 1000 on $\mathcal{P}_{\text{val}}$.
3) Select the earliest checkpoint attaining the minimum validation loss. The rail is
$$\mathcal{K} = \{0, 25, \ldots, 1000\}; \tag{S11}$$
ties therefore retain the smaller update count.

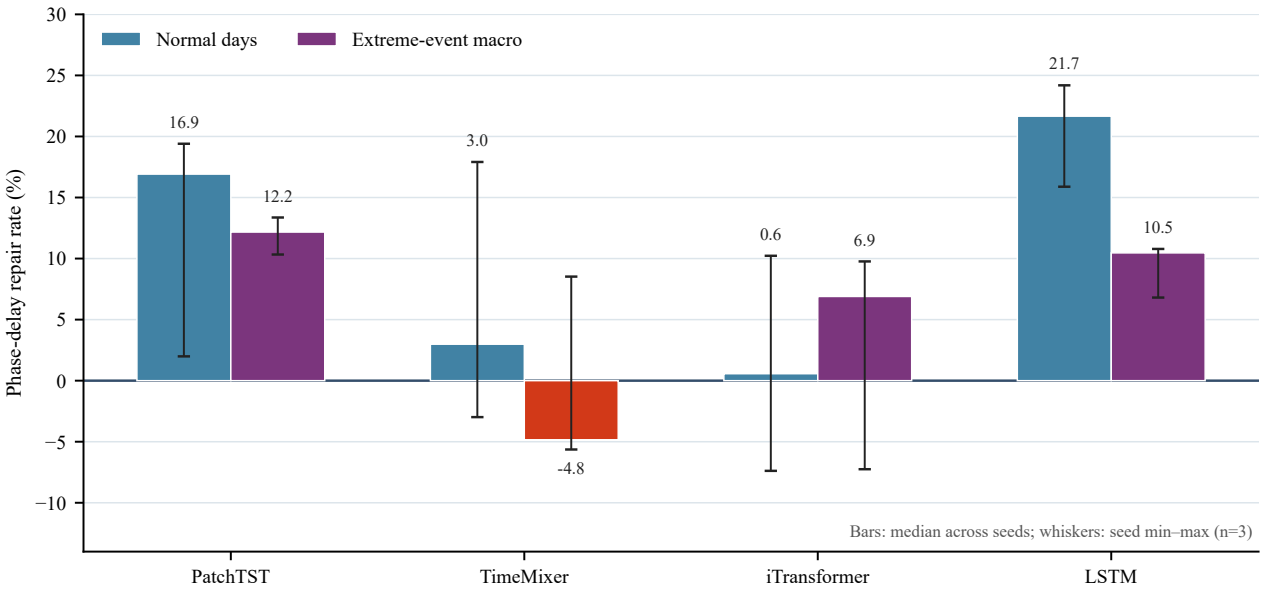


Fig. S7. Phase-delay repair in the Shandong validation period across four frozen base forecasters. Blue: normal days; violet/red: the equally weighted high-price/negative-price event macro, with red denoting negative median improvement. Bars are three-seed medians and whiskers are seed ranges. Evaluation requires revealed outcome timestamps and does not enter HCH's issued-time decision.

4) Recreate the model from the identical initialization θ_0 and fit the complete legal prefix $\mathcal{P}$ for exactly the selected number of updates. This refit becomes the frozen direction estimator used by later OOF calibration and deployment.

CACC changes effective fitting capacity only. It adds no trainable parameters and never reads the target-day outcome.

Implementation and reproducibility. The direction estimator projects seven historical daily patches and the current context into 32-dimensional representations, processes them with a single-layer GRU of hidden width 32, and maps the output to the 24-hour signed direction; HCRDE uses no dropout. Full-batch direction fitting uses AdamW with learning rate 10^{-3}, weight decay 10^{-4}, and gradient-norm clipping at 1.0. The deployed corrector requires the issued Base forecast, forecast-known state, four deterministic calendar features per horizon position, and revealed historical prices/residuals, without using the current target-day outcome.

The principal environment reported in the main paper is Ubuntu 22.04, Python 3.10, PyTorch 2.1.2, CUDA 11.8, and an NVIDIA RTX 4090D (24 GB). CPU-only accuracy replays and GPU runtime/memory profiling are distinct measurements. The accompanying code repository documents dependency installation and executable `fit`, `predict`, and `score` commands with separate issued-cache and outcome files; complete market-level replay additionally requires properly licensed frozen Base forecasts and issue-time covariate records.